

L 11 : PROBABILITIES (PART II)

Using Tree Diagram to list out the possible outcomes

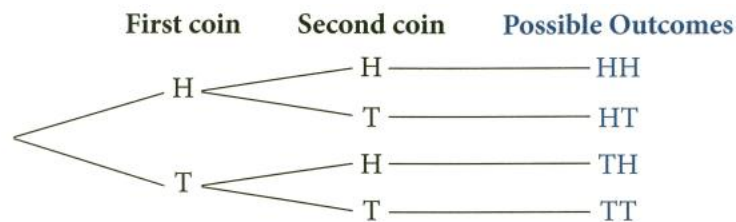
When we toss **1 coin**, the possible outcomes are 'Head' and 'Tail' only.

When we toss **2 coins**, there would be more possible outcomes.
(Different combination of 'Head' and 'Tail'.)



In this case, we would use a **Tree diagram** to list out all the possible outcomes.

'H' stands for 'Head' while 'T' stands for 'Tail'



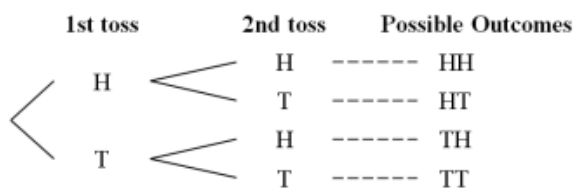
From the tree diagram, there are **4 possible outcomes**, when we toss 2 coins.

Example : $P(\text{getting exactly 1 head and 1 tail}) = \frac{2}{4} = \frac{1}{2}$

1. Candy tosses a fair coin twice.

- List all the possible outcomes by a tree diagram.
- Find the probability that Candy gets two tails.

(a) Let H stand for a head and T stand for a tail.

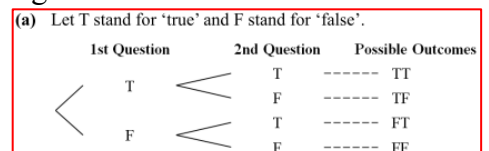


(b) From the tree diagram, there are 4 equally likely outcomes.
There is only 1 favourable outcome, i.e. TT.

$$\therefore P(2 \text{ tails}) = \frac{1}{4}$$

2. There are 2 true-or-false questions in a quiz. The correct answers to the two questions are 'false' and 'true' respectively. Ben answers the 2 questions by wild guessing.

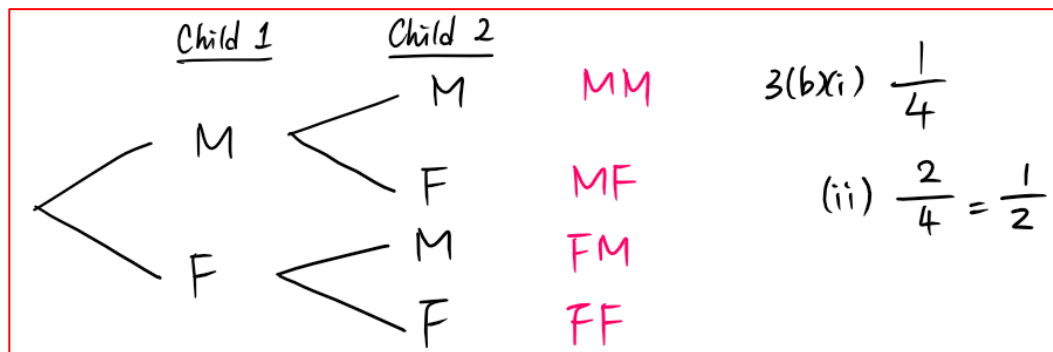
- List all the possible outcomes by a tree diagram.
- Find the probability that he gets only one answer correct.



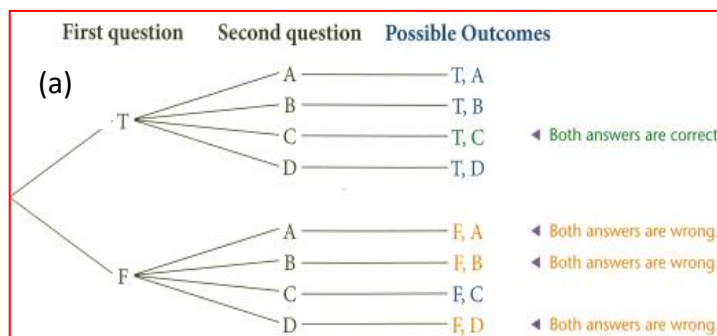
(b) From the tree diagram, there are 4 equally likely outcomes.
There are 2 favourable outcomes, i.e. TT, FF.

$$\therefore P(\text{only one answer correct}) = \frac{2}{4} = \frac{1}{2}$$

3. Mr. Chan has two children.
- List all the possible outcome by a tree diagram.
 - Find the probabilities that he has
 - two daughters
 - 1 daughter only



4. There are two questions in a quiz. The first one is a true-or-false question and the answer is 'true'. The second one is a multiple-choice question which offers 4 choices: A, B, C, D and the answer is 'C'. If Wallace chooses his answers at random,
- List all the possible outcomes by a tree diagram.
 - find the probabilities that he gets
 - both answers correct,
 - both answers wrong.



From the tree diagram, total number of equally likely outcomes = 8.

- (b)(i) There is only 1 favourable outcome, i.e. T, C.

$$\therefore P(\text{both answers correct}) = \frac{1}{8}$$

- (b)(ii) There are 3 favourable outcomes, i.e. F, A; F, B; F, D.

$$\therefore P(\text{both answers wrong}) = \frac{3}{8}$$

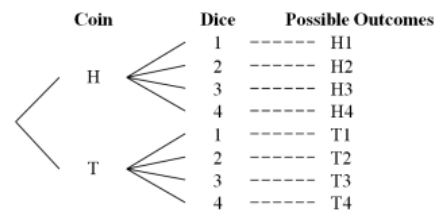
5. A coin is tossed and a dice in the shape of a regular tetrahedron is thrown. The numbers 1, 2, 3 and 4 are marked near the vertices of the dice. The outcome of throwing this dice is the number near to top of the dice.

- (a) List all the possible outcomes by a tree diagram.
(b) Find the probabilities of getting
(i) a head and the number '3',
(ii) a tail and a prime number.

(ii) There are 2 favourable outcomes, i.e. T2 and T3.

$$\begin{aligned}\therefore P(\text{a tail and a prime number}) &= \frac{2}{8} \\ &= \frac{1}{4}\end{aligned}$$

(a) Let H stand for a head and T stand for a tail.



(b) (i) From the tree diagram, total number of possible outcomes = 8
There is only 1 favourable outcome, i.e. H3.

$$\therefore P(\text{a head and the number '3'}) = \frac{1}{8}$$

Using Tabulation to list out the possible outcome

Besides using tree diagrams, we may also list all the possible outcomes in tables. For example, the table as shown lists all the possible outcomes of tossing a coin twice, where H stands for a head and T stands for a tail.

		Second toss	
		H	T
First toss	H	HH	HT
	T	TH	TT

6. Two dice are thrown.

- (a) List all the possible outcomes in a table.
(b) Find the probabilities that
(i) the difference of the 2 numbers is equal to 3,
(ii) the sum of the 2 numbers is a multiple of 3.

(b) From the table, there are 36 possible outcomes altogether.

(i) There are 6 outcomes in which the difference of the 2 numbers is equal to 3.

$$\begin{aligned}\therefore P(\text{difference equal to 3}) &= \frac{6}{36} \\ &= \frac{1}{6}\end{aligned}$$

(ii) There are 12 outcomes in which the sum of the 2 numbers is a multiple of 3.

$$\begin{aligned}\therefore P(\text{sum equal to a multiple of 3}) &= \frac{12}{36} \\ &= \frac{1}{3}\end{aligned}$$

		Second outcome					
		1	2	3	4	5	6
1st outcome	1						
	2						
	3						

(a)

		2nd dice					
		1	2	3	4	5	6
1st dice	1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
	2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
	3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
	4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
	5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
	6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

7. Edith selects two numbers at random from 1, 2 and 4 to form a 2-digit number. These numbers can be selected more than once.

- (a) Use a table to list all the possible 2-digit numbers.
- (b) Find the probabilities that the 2-digit number formed is
 - (i) a multiple of 3,
 - (ii) divisible by 4.

(b) From the tree diagram, total number of possible outcomes = 9

(i) There are 4 favourable outcomes, i.e. 12, 21, 24, 42.

$$\therefore P(\text{a multiple of 3}) = \frac{4}{9}$$

(ii) There are 3 favourable outcomes, i.e. 12, 24, 44.

$$\therefore P(\text{divisible by 4}) = \frac{3}{9} = \frac{1}{3}$$

(a)

		Unit digit		
		1	2	4
Tenth digit	1	11	12	14
	2	21	22	24
	4	41	42	44

8. A shop offers a gift-wrapping service as shown below.

Sara has bought a gift from the shop. She needs a piece of wrapping paper and a ribbon for the wrappings.

- (a) With the help of a table, list all the possible outcomes of the wrapping cost of Sara's gift.
- (b) Find the probability that she pays more than \$15 for the wrappings.

For each gift, please choose a piece of wrapping paper and a piece of ribbon.

Wrapping paper		Ribbon	
Red	\$10	Green	\$8
Yellow	\$8	Red	\$12
Blue	\$9	Yellow	\$4

(a) Let R, Y, B and G stand for red, yellow, blue and green respectively.

		Ribbon		
		G	R	Y
Wrapping paper	R	RG (\$18)	RR (\$22)	RY (\$14)
	Y	YG (\$16)	YR (\$20)	YY (\$12)
	B	BG (\$17)	BR (\$21)	BY (\$13)

(b) From the table, number of possible outcomes = 9

There are 6 outcomes in which Sara pays more than \$15 for the wrappings.

$$\therefore P(\text{more than \$15}) = \frac{6}{9} = \frac{2}{3}$$

PRE S3 MATHS L11 – PROBABILITY (II)

9. A pair of white socks, a pair of black socks and a pair of red socks are put in a drawer. Jason draws 2 socks one by one from the drawer at random without replacement.

- (a) Use a table to list all the possible outcomes.
(b) Find the probabilities that
(i) the 2 socks are of the same colour,
(ii) at least one of the socks is black in colour.

(a) Let (B_1, B_2) , (W_1, W_2) and (R_1, R_2) represent the 3 pairs of socks respectively.

	2nd sock					
	B_1	B_2	W_1	W_2	R_1	R_2
1st sock	B_1	B_2	W_1	W_2	R_1	R_2
B_2	B_2B_1		B_2W_1	B_2W_2	B_2R_1	B_2R_2
W_1	W_1B_1	W_1B_2		W_1W_2	W_1R_1	W_1R_2
W_2	W_2B_1	W_2B_2	W_2W_1		W_2R_1	W_2R_2
R_1	R_1B_1	R_1B_2	R_1W_1	R_1W_2		R_1R_2
R_2	R_2B_1	R_2B_2	R_2W_1	R_2W_2	R_2R_1	

(b)(i) There are 6 outcomes in which the 2 socks are of the same colour.

$$\therefore P(2 \text{ socks of the same colour}) = \frac{6}{30} = \frac{1}{5}$$

(b)(ii) There are 18 outcomes in which at least one of the socks is black in colour.

$$\therefore P(\text{at least one of the socks is black in colour}) = \frac{18}{30} = \frac{3}{5}$$

10. A bag contains 1 green marble, 1 black marble and 2 red marbles. Two marbles are drawn successively from the bag at random.

- (a) List all the possible outcomes in a table.
(Use G to denote a green marble, B to denote a black marble and R to denote a red marble.)
(b) Find the probability that the two marbles drawn are of different colours.

(a)

	2nd marble drawn			
	G	B	R	R
1st marble drawn	G	G, B	G, R	G, R
B	B, G		B, R	B, R
R	R, G	R, B		R, R
R	R, G	R, B	R, R	

(b) From the table,
total number of possible outcomes = 12
There are 10 favourable outcomes.

$$\therefore P(\text{different colours}) = \frac{10}{12} = \frac{5}{6}$$

11. There are 3 men and 2 women in a tennis class. If 2 persons are chosen at random to play a match,

- (a) Use a table to list all the possible outcomes.
(b) find the probabilities that the 2 persons are
(i) 2 men,
(ii) 1 man and 1 woman.

From the table, there are 20 possible outcomes altogether.

(a) There are 6 outcomes in which 2 men are chosen.

$$\therefore P(2 \text{ men}) = \frac{6}{20} = \frac{3}{10}$$

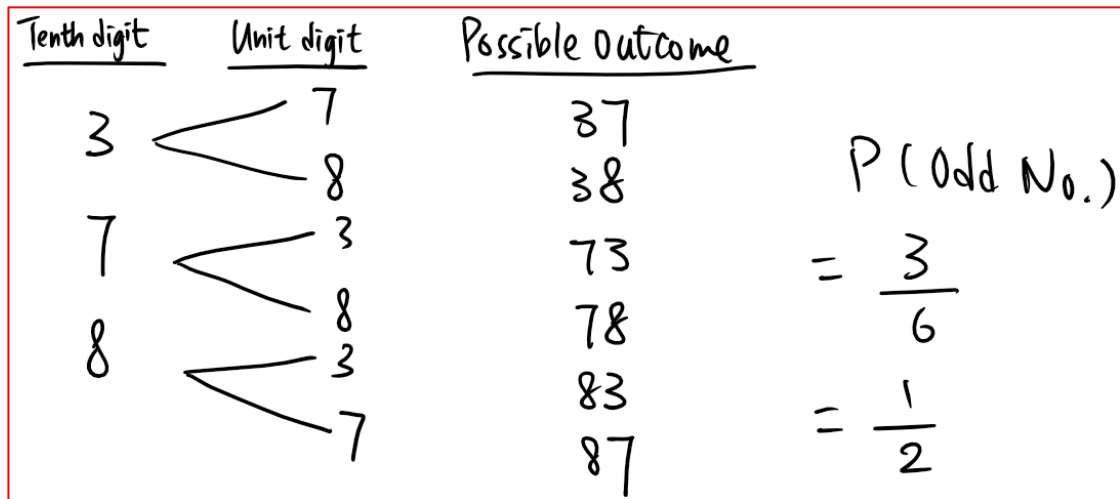
(b) There are 12 outcomes in which 1 man and 1 woman are chosen.

$$\therefore P(1 \text{ man and 1 woman}) = \frac{12}{20} = \frac{3}{5}$$

	Second person				
	M_1	M_2	M_3	W_1	W_2
First person	M_1	M_1M_2	M_1M_3	M_1W_1	M_1W_2
M_2	M_2M_1		M_2M_3	M_2W_1	M_2W_2
M_3	M_3M_1	M_3M_2		M_3W_1	M_3W_2
W_1	W_1M_1	W_1M_2	W_1M_3		W_1W_2
W_2	W_2M_1	W_2M_2	W_2M_3	W_2W_1	

Application Questions

12. Two numbers are chosen at random one by one from the numbers '3, 7 and 8'. A number is formed by placing the first drawn number as the tens digit and the second drawn number as the unit digit. What is the probability that the number formed is an odd number?



13. Celine has a \$10 note, a \$20 note, a \$50 note and a \$100 note in her purse. If she takes out 2 notes randomly from her purse, find the probability that she gets at least \$70.

1st note	2nd note	Possible Outcomes
\$10	\$20	\$(10 + 20) = \$30
	\$50	\$(10 + 50) = \$60
	\$100	\$(10 + 100) = \$110
\$20	\$10	\$(20 + 10) = \$30
	\$50	\$(20 + 50) = \$70
	\$100	\$(20 + 100) = \$120
\$50	\$10	\$(50 + 10) = \$60
	\$20	\$(50 + 20) = \$70
	\$100	\$(50 + 100) = \$150
\$100	\$10	\$(100 + 10) = \$110
	\$20	\$(100 + 20) = \$120
	\$50	\$(100 + 50) = \$150

From the tree diagram, total number of possible outcomes = 12

Number of favourable outcomes = 8

$$\therefore P(\text{at least } \$70) = \frac{8}{12} = \frac{2}{3}$$

14. There are four gateways A, B, C and D in an exhibition hall. Josephine enters through one of the four gateways at random and leaves through one of the other three gateways at random. Find the probability that Josephine didn't pass through gateway D.

$$(a) P(\text{didn't pass D}) = \frac{6}{12} \\ = \frac{1}{2}$$

		Entering through			
		A	B	C	D
Leave through	A		(B,A) (C,A) (D,A)		
	B	(A,B)		(C,B) (D,B)	
	C	(A,C) (B,C)			(D,C)
	D	(A,D) (B,D) (C,D)			

15. Rosa arranges the numbers '6', '5' and '1' randomly to form a 3-digit number. Each number can be used once only. Find the probability that the number formed is a multiple of 5.

Hundreds digit	Tens digit	Units digit	Possible Outcomes
6	5	1	----- 651
	1	5	----- 615
5	6	1	----- 561
	1	6	----- 516
1	6	5	----- 165
	5	6	----- 156

From the tree diagram, total number of possible outcomes = 6
 Number of favourable outcomes = 2 (i.e. 615 and 165)

$$\therefore P(\text{a multiple of 5}) = \frac{2}{6} \\ = \frac{1}{3}$$

PRE S3 MATHS L11 - PROBABILITY (II)

16. In bag A, there are 1 green ball and 2 yellow balls and 1 red ball. In bag B, there are 2 green balls and 1 yellow ball. Ball X is randomly drawn from bag A and put into bag B.
- (a) Find the probability that ball X is not yellow.
- (b) After ball X is put into bag B, a ball Y randomly drawn from bag B. Note that ball X and ball Y may refer to the same ball.
- (i) If ball X is yellow, find the probability that ball Y is red.
- (ii) Find the probability that ball Y is red.

Q16 (a) $\frac{1}{2}$

(b)(i) 0

(ii)

<u>Ball X</u> Taken out from bag A, put into bag B	<u>Ball Y</u> Taken out from bag B
Green	Green Green Green Yellow
Yellow(1)	Green Green Yellow Yellow
Yellow(2)	Green Green Yellow Yellow
Red	Green Green Yellow Red

∴ The required probability is $\frac{1}{16}$ //